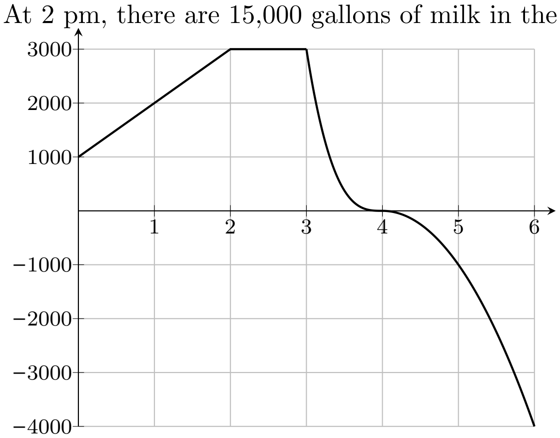
Class 29

The Fundamental Theorem of Calculus, Part 1

Today's Goals

- You should be able to visualize accumulation functions of the form $G(x) = \int_a^x f(t)\,dt$ in terms of signed area and interpret them in terms of net change.
- You should be able to apply part 1 of the Fundamental Theorem of Calculus, and explain why it makes sense when f is a rate function.
- You should be able to sketch a graph of $G(x) = \int_a^x f(t)\,dt$ when given a formula or graph of f.

1. At the Blue Bell ice cream factory, milk is stored in a 40,000 gallon tank. Below is the graph of $f(t)$, the rate (in gallons per hour) at which milk is flowing into and out of the tank t hours after noon one day. At 2 pm, there are 15,000 gallons of milk in the tank.



(a) How much milk is in the tank at 3 pm?

Solution

Since there are 15,000 gallons of milk in the tank at 2 pm, the amount of milk in the tank at 3 pm is $15,000 + \int_2^3 f(t)\,dt = 15,000 + 3,000 = \boxed{18,000 \text{ gallons}}$.

(b) How much milk is in the tank at noon?

Solution

Since there are 15,000 gallons of milk in the tank at 2 pm, the amount of milk in the tank at noon is $15,000 + \int_2^0 f(t)\,dt = 15,000 - \int_0^2 f(t)\,dt = 15,000 - 4,000 = \boxed{11,000 \text{ gallons}}$.

(c) Write an expression giving the amount of milk in the tank x hours after noon.

Solution

$15,000 + \int_2^x f(t)\,dt.$

(d) Let $M(x)$ be the amount of milk in the tank x hours after noon. What is $M'(5)$?

Solution

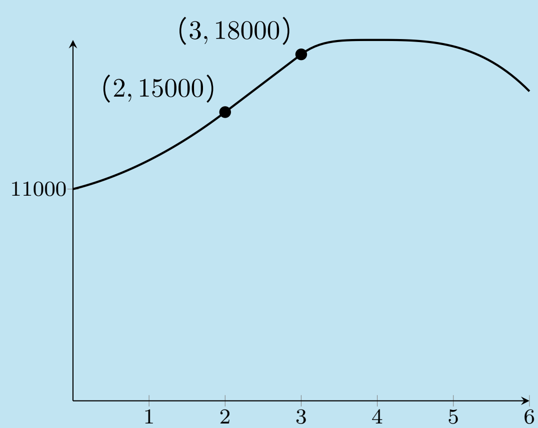
$M'(5)$ is the instantaneous rate of change in the amount of milk at 5 pm, which is exactly $f(5) = \boxed{-1000}$ gallons per hour.

(e) Sketch the graph of M.

Solution

The derivative of M is exactly the rate graph we're given (that is, $M' = f$), so we basically want to sketch M from knowing the graph of M'. We also know that $M(2) = 15,000$.

On $(0,2)$, $M' = f$ is positive and increasing, so M is increasing and concave up. On $(2,3)$, $M' = f = 3000$, so M is linear with slope 3000. On $(3,4)$, $M' = f$ is positive and decreasing, so M is increasing and concave down. Finally, on $(4,6)$, $M' = f$ is negative and decreasing, so M is decreasing and concave down. We also found already that $M(3) = 18,000$ and $M(0) = 11,000$. So, here's the graph of M:

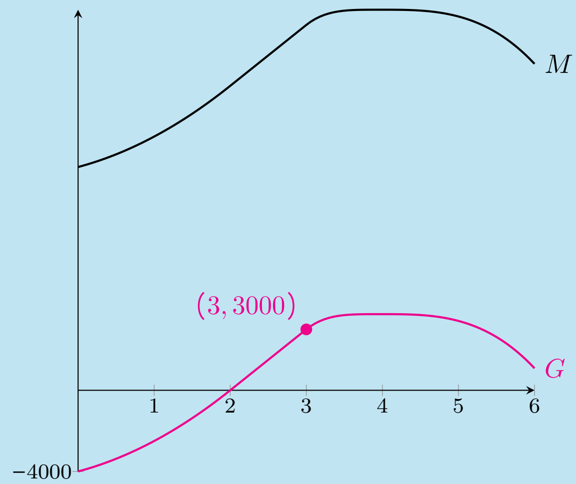


(Around $t = 4$, f is close to 0, so the graph of M is pretty flat.)

(f) Let $G(x) = \int_2^x f(t)\,dt$. In words, what does $G(x)$ represent? Sketch the graph of G on your graph from (e).

Solution

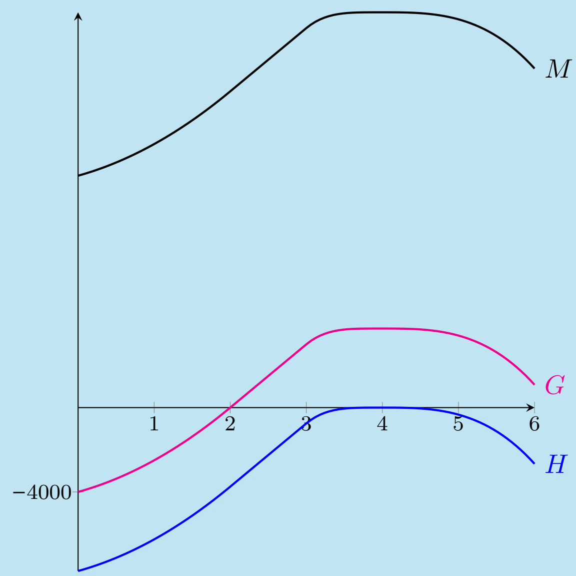
$G(x)$ represents the net change in amount of milk from time 2 to time x, or (amount of milk at time x) − (amount of milk at time 2). This is the same as $M(x) - 15000$, so to graph G, we just shift the graph of M down by 15000:



(g) Let $H(x) = \int_4^x f(t)\,dt$. In words, what does $H(x)$ represent? Sketch the graph of H on your graph in (e).

Solution

$\int_4^x f(t)\,dt$ is the net change in amount of milk from time 4 to time x, or (amount of milk at time x) − (amount of milk at time 4), or $M(x) - M(4)$. So, to graph H, we shift the graph of M down by $M(4)$ (which means the new graph has an x-intercept at $x = 4$):



(h) Find $G'(5)$ and $H'(5)$. (Include units.)

Solution

We already said that $M'(5) = f(5) = -1000$ gallons per hour. Since G and H are just vertical shifts of M, they have the same slope at 5; that is,

$$\boxed{G'(5) = H'(5) = M'(5) = -1000 \text{ gallons per hour}}.$$

(i) **Summarize:** What are the important relationships among f, M, G, and H?

Solution

First, M', G', and H' are equal to f. Second, M, G, and H are vertical shifts of each other.

The Fundamental Theorem of Calculus, Part 1 (FTOC 1)

If f is a continuous function and a is a number, let $G(x) = \int_a^x f(t)\,dt$. Then $G'(x) = \underline{\boldsymbol{f(x)}}$.

2. (a) If $S(x) = \int_3^x \sin\left(\frac{\pi t^2}{2}\right) dt$, what is $S'(4)$?

Solution

If we write $f(t) = \sin\left(\frac{\pi t^2}{2}\right)$, then FTOC 1 says that $G'(x) = f(x)$. Therefore, $G'(4) = f(4) = \sin(8\pi) = \boxed{0}$.

(b) Find $\frac{d}{dx}\left[\int_3^x \sin\left(\frac{\pi t^2}{2}\right) dt\right]$.

Solution

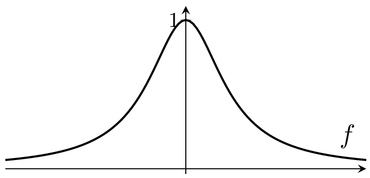
Using the same notation as in the previous part, this is asking for $G'(x)$, which is simply $f(x) = \boxed{\sin\left(\frac{\pi x^2}{2}\right)}$ by FTOC 1.

(c) Find $\frac{d}{dx}\left(\int_a^b f(x)\,dx\right)$, where f is a continuous function on $[a, b]$.

Solution

$\int_a^b f(x)\,dx$ is a constant, so its derivative is $\boxed{0}$. (Note that we generally consider $\frac{d}{dx}\left(\int_a^b f(x)\,dx\right)$ to be bad notation; it's not incorrect, but it's confusing: the $\frac{d}{dx}$ tells us we're thinking of x as a variable, so it's better to use a different variable as the "dummy variable" in our integral. So, it's better to write $\frac{d}{dx}\left(\int_a^b f(t)\,dt\right)$ or $\frac{d}{dx}\left(\int_a^b f(s)\,ds\right)$ to avoid confusion.)

3. The graph of a function $f(x)$ is shown below. Let $G(x) = \int_3^x f(t)\,dt$. Answer the following questions about G, using FTOC 1 as appropriate.



(a) What is $G(3)$?

Solution

By definition, $G(3) = \int_3^3 f(t)\,dt$, and this integral is 0.

(b) On what intervals is G increasing? Decreasing? How do you know?

Solution

G is increasing when $G' > 0$ and decreasing when $G' < 0$. By Part 1 of FTOC, $G' = f$. So, the previous sentence can be rewritten to say, "G is increasing when $f > 0$ and decreasing when $f < 0$." In this case, $f(x)$ is always positive, so that means that G is always increasing; that is, $\boxed{G \text{ is increasing on } (-\infty, \infty)}$.

(c) On what intervals is G concave up? Concave down?

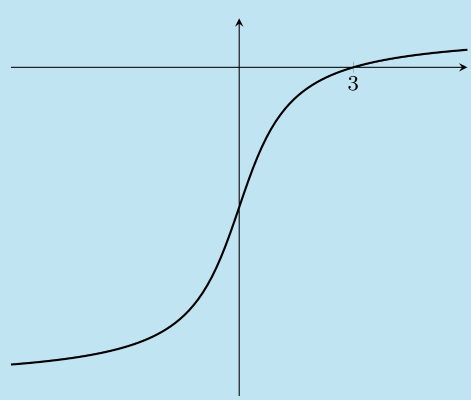
Solution

G is concave up when $G'' > 0$, and G is concave down when $G'' < 0$. By Part 1 of FTOC, $G' = f$. So, the previous sentence can be rewritten to say "G is concave up when $f' > 0$ (in other words, when f is increasing) and G is concave down when $f' < 0$ (in other words, when f is decreasing)." In this case, f is increasing when $x < 0$ and decreasing when $x > 0$. So, $\boxed{G \text{ is concave up on } (-\infty, 0) \text{ and concave down on } (0, \infty)}$.

(d) Sketch the graph of G.

Solution

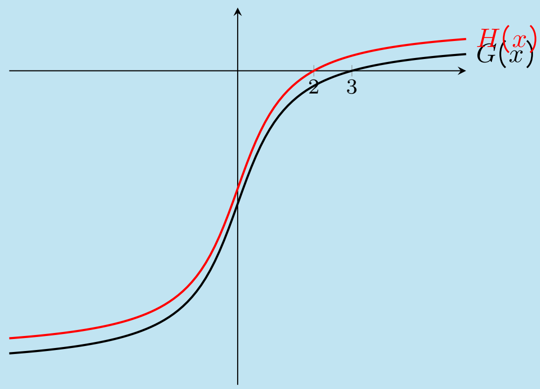
Using our answers to the previous parts, G should look something like this:



(e) Let $H(x) = \int_2^x f(t)\,dt$. Sketch the graph of H on your graph in (d). How does G relate to H?

Solution

By FTOC 1, $H' = f$, so G and H have the same derivative. Therefore, they are vertical shifts of each other. Since $H(2) = 0$, we can get the graph of H by shifting the graph of G vertically until it passes through $(2, 0)$:



4. The rate that water is entering/leaving a tank is given by $f(t) = 60 - 3t$ gallons/minute, where t is measured in minutes past noon. Let $F(x) = \int_0^x f(t)\,dt$.

(a) Interpret $F(x)$ in words, for $x \geq 0$.

Solution

$F(x) = \int_0^x f(t)\,dt$ is the net change in the amount of water in the tank between noon and x minutes after noon. In other words, it is (amount of water in the tank x minutes after noon) $-$ (amount of water in the tank at noon).

(b) For what values of x is $F(x)$ increasing?

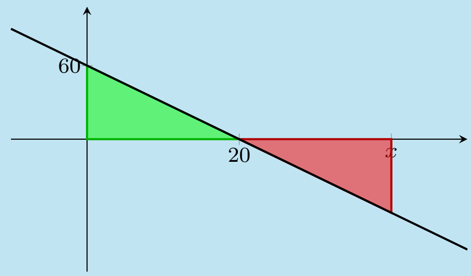
Solution

$F(x)$ is increasing when $F'(x) = f(x) = 60 - 3x$ is positive, which happens for $\boxed{x < 20}$.

(c) At what time is the water level in the tank the same as it was at noon? At this time, what is the value of $F(x)$?

Solution

We are looking for a time x when the net change in amount of water from noon to x is 0; that is, we want $\int_0^x f(t)\,dt = 0$. Graphically, we want to find the value of x so that the signed area under $f(t)$ from $t = 0$ to $t = x$ is 0:

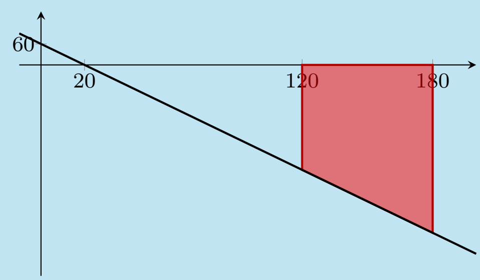


By symmetry, we can see that $x = 40$, which corresponds to $\boxed{\text{12:40 pm}}$.

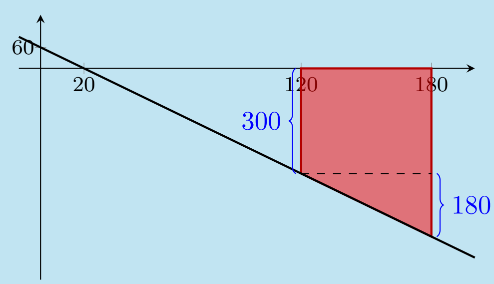
(d) If the water tank has 30,000 gallons of water at 2 pm, how much does it have at 3 pm? (Be careful with units!)

Solution

The net change in the amount of water from 2 pm ($t = 120$) to 3 pm ($t = 180$) is $\int_{120}^{180} f(t)\,dt$. We can picture this graphically as the following signed area:



Since $f(120) = -300$ and $f(180) = -480$, we can break this trapezoid up into a rectangle of height 300 and a triangle of height 180:



The rectangle and triangle both have width 60, so the total area of the trapezoid is $60 \cdot 300 + \frac{1}{2} \cdot 60 \cdot 180 = 23400$. Therefore, the net change in the amount of water from 2 pm to 3 pm is -23400 gallons. Since there were 30000 gallons of water at 2 pm, the amount of water at 3 pm was $30000 - 23400 = \boxed{\text{6600 gallons}}$.